Reachable Nodes

Time limit: 1.00 s **Memory limit:** 512 MB

A directed acyclic graph consists of n nodes and m edges. The nodes are numbered $1, 2, \ldots, n$.
Calculate for each node the number of nodes you can reach from that node (including the node itself).

Input

The first input line has two integers n and m: the number of nodes and edges.

Then there are m lines describing the edges. Each line has two distinct integers a and b: there is an edge from node a to node b.

Output

Print n integers: for each node the number of reachable nodes.

Constraints

- $1 \leq n \leq 5 \cdot 10^4$
- $1 \leq m \leq 10^5$

Example

Input	Output
5 6 1 2 1 3 1 4 2 3 3 5 4 5	5 3 2 2 1